

Catch-Rematched Hybrid Flare-Gated Transit: A Packet-Centered Active-Rail Architecture for Source-Supplied Spacetime Engineering

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Abstract

This paper presents a standalone reduced-engineering reference model for a composite faster-than-light spacetime architecture under the source-available assumption. The model combines hybrid flare-gated wormhole infrastructure with a catch-rematched throat-loaded transit packet. The core result is a design architecture: passenger safety, infrastructure accessibility, transition-current burden, and spatial-curvature burden become separate engineering channels. The frozen v1 candidate treats the throat as active rail infrastructure and the passenger as a protected moving packet. The packet is caught before shift fade, shift support fades before throat relaxation, and the shift remains contained inside the same throat support envelope that supplies lapse and capacity. The frozen balanced default uses

$C_0 = 20$, $C_\perp = 5$, $R_{\text{th}} = 1.25$, $w_{\text{th}} = 0.12$, $R_{\text{pass}} = 0.35$, $w_{\text{pass}} = 0.08$, $B_0 = 6$, $w_B = 12$,
with

$x_{\text{catch}} = 0.05$, $x_\beta = 0.70$, $x_q = 1.25$, $w_{\text{catch}} = 0.25$, $w_\beta = 0.28$, $w_q = 0.30$, $p_\beta = 4$.

The reduced ADM screens show that transition profiles govern the dynamic/current channel (j_M, K), while capacity/support geometry governs the Hamiltonian/spatial-curvature channel ($\rho_H, {}^{(3)}R$). The three-scenario geometry confirmation reduces total source-demand proxy to about $0.162\times$ the v01 geometry baseline, geometry cost to $0.274\times$, and dynamic cost to $0.0108\times$. The conservative current-floor source screen reduces Lorentzian-sampled packet negative exposure at $V = 10$, $\lambda = 6$, and $\tau = 0.2$ to about $0.042\times$ baseline. Robustness checks identify a mixed-capacity operating band, with $C_\perp = 5$ as the balanced default and $C_\perp = 3$ as a pressure/current-leaning branch. The paper's central contribution is a burden-distribution map for source-supplied spacetime transport: the passenger rides a packet, the throat acts as plant, transition profiles manage current, and angular capacity manages spatial-curvature concentration.

1 The engineering object

The usual language of traversable wormholes and warp drives begins with a shape. A throat connects two regions. A bubble moves through an exterior. A route is prepared. These shapes are the right starting point for geometry. A transport system needs one more layer of description. It needs a service object.

A passenger system is a sequence of controlled events. Support is prepared. Capacity is supplied. Motion is coupled. The passenger is caught. The transport field is faded. The infrastructure relaxes and resets. The engineering object is the whole service process together with the metric at each instant.

The combined architecture developed here takes that service view seriously. The wormhole throat is active infrastructure. It stretches, opens, gates, catches, supports, and relaxes. The passenger is represented by a finite packet. The packet is the cabin-level object; the throat is the plant. The transport field is the controlled coupling between the two.

This framing changes the technical question. A passive tunnel design asks for a broadly comfortable throat. A standalone bubble design asks the moving wall to carry both transport and passenger protection. The composite design asks how burden should be distributed across a source-supplied plant:

$$\boxed{\text{packet safety} + \text{transition current control} + \text{capacity support} + \text{spatial-curvature management} + \text{infrastructure re}} \quad (1)$$

The frozen v1 candidate is the first reduced model in this line that makes that distribution explicit.

2 The contribution in engineering terms

The paper contributes four concrete design results.

First, it separates passenger safety from infrastructure accessibility. The passenger criterion is the moving packet worldtube. Stationary g_{tt} remains an infrastructure and station-access diagnostic. This separation lets the model evaluate what the passenger actually rides while treating passive comfort throughout the active plant as an infrastructure question.

Second, it identifies catch/rematch choreography as the hard packet gate. The service order

$$\boxed{\text{catch} \rightarrow \text{shift fade} \rightarrow \text{throat relaxation}} \quad (2)$$

keeps the packet supported through the shift transition. Late catch produces packet failure because the still-fast packet loses the support field that carried it.

Third, it separates burden channels. Transition profiles move j_M and K . Capacity/support geometry moves ρ_H and ${}^{(3)}R$. This turns a broad exotic-source burden into two engineering subsystems.

Fourth, it identifies mixed radial/angular capacity as the spatial-burden solution in the tested composite family. The angular capacity $C_\perp$ is a curvature-management control. The useful range is mixed-capacity, with $C_\perp = 5$ as the balanced default and $C_\perp = 3$ as a pressure/current-leaning branch.

Together these results turn the combined wormhole/warp construction into a source-supplied service architecture. The literature supplies the physical warning signs: exotic support, thin-wall severity, length-scale pressure, capacity bookkeeping, prepared infrastructure, and ADM role separation. The present result shows how those pressures distribute when the system is engineered as active throat infrastructure carrying a protected packet.

3 What the literature makes clear, and what this model adds

The field already provides the boundary conditions for any serious FTL geometry. Morris and Thorne establish the traversable throat as a geometric object with flare-out, finite redshift, low tidal-force, and exotic-support requirements [1]. Morris, Thorne, and Yurtsever connect wormhole

mouth arrangements to chronology control [2]. Ford and Roman show that quantum inequalities turn negative energy into a magnitude-duration and length-scale problem [3]; Fewster and Roman sharpen the null-contracted sampled-stress pressure for small-exotic-matter proposals [4]. Dynamic-throat work keeps local null expansion and local energy-condition diagnostics at the throat [5].

Warp and prepared-route papers supply complementary pieces. Alcubierre shows how a shift-dominated warp metric can be written in general relativity [6]. Pfenning and Ford show how the Alcubierre wall concentrates severe quantum-inequality pressure [7]. Van den Broeck shows that spatial capacity can change energy accounting by separating compact exterior size from interior volume [8]. Natario emphasizes the shift-structure view of warp motion beyond expansion/contraction intuition [9]. Everett and Roman’s Krasnikov-tube analysis places prepared infrastructure and causal control at the center of route-based superluminal travel [10]. Bobrick and Martire provide an ADM language in which lapse, shift, spatial geometry, capacity, and time-rate roles can be separated [11]. Lentz and Fuchs et al. show the continuing value of matter-shell and shift-structure engineering in modern warp-geometry work [12, 13]. Gao, Jafferis, and Wall show controlled traversability through source timing/coupling in a holographic setting [14].

Those papers make the stakes legible. They leave an engineering allocation problem. If source support is available, which part of the geometry should carry passenger safety? Which part should carry transition current? Which part should carry spatial curvature? Which field should behave like a rail, which like a cabin, which like a switch, and which like a reset mechanism?

The answer developed here is summarized in Table 1. The table is the paper’s literature integration in compact form: broad field principle on the left, composite-specific result on the right.

Table 1: Field principles and composite-design clarifications.

Field principle	Composite-design clarification
Traversable throats need flare-out support and passenger-compatible access.	The throat functions as active infrastructure. Passenger compatibility attaches to a protected moving packet worldtube.
Quantum inequalities punish concentrated negative energy and thin support layers.	Broadened support and mixed radial/angular capacity reduce reduced ADM source-demand and conservative packet source-exposure proxies.
Warp metrics separate lapse, shift, and spatial geometry in ADM form.	The composite screen assigns burdens: transition profiles control j_M, K ; capacity/support geometry controls $\rho_H, {}^{(3)}R$.
Capacity engineering changes accounting.	Capacity splits into a radial service channel C_0 and an angular curvature-relief channel $C_\perp$.
Prepared infrastructure can carry route burden.	The throat gates the shift and supplies the capacity/lapse envelope: $\text{supp}(\beta^l) \subseteq \text{supp}(A, T)$.
Source timing can open traversable channels in special settings.	Packet service uses a concrete timing basin: catch, shift fade, throat relaxation.

4 Architecture: the throat becomes plant

4.1 Hybrid flare-gated infrastructure

The throat infrastructure begins with the hybrid flare-gated form

$$ds^2 = -N(l, t)^2 dt^2 + B(l, t)^2 dl^2 + R(l, t)^2 d\Omega^2. \quad (3)$$

The control roles are direct:

$B(l, t)$: radial proper-length support and service-coordinate stretch,

$R(l, t)$: flare/access state and angular geometry,

$N(l, t)$: timing, lapse margin, and shoulder matching.

The wormhole-only v1 model used these functions to create a phased throat-support plant: prestretch, flare opening, quiet access hold, closure, repayment/shoulder matching, and reset [15]. The composite model uses the same controls with a new job. They carry the infrastructure around a service packet.

4.2 Throat-loaded packet

The packet side is written in radial ADM form,

$$ds^2 = -T^2 dt^2 + A^2 (dl + \beta^l dt)^2 + A^2 r(l)^2 d\Omega^2. \quad (4)$$

Capacity and lapse are paired,

$$A = \exp(qW \ln C_0), \quad T = \exp(qW \ln(\lambda C_0)), \quad (5)$$

so that

$$T/A = \lambda^{qW}. \quad (6)$$

The shift is throat-gated:

$$\beta^l(l, t) = -U(t)E(X(t))W(l)S(l - X(t)). \quad (7)$$

The support-inclusion rule

$$\text{supp}(\beta^l) \subseteq \text{supp}(A, T) \quad (8)$$

keeps the transport shift inside the same region that supplies support. This rule is one of the bridge results from the original catch-rematched transit ansatz [16].

4.3 Final composite field

The final composite metric is

$$ds^2 = -\alpha(l, t)^2 dt^2 + \gamma_{ll}(l, t)(dl + \beta^l(l, t)dt)^2 + \gamma_{\Omega\Omega}(l, t)d\Omega^2. \quad (9)$$

The fields are

$$\alpha(l, t) = N_{v1}(l, t)T_{\text{pkt}}(l, t), \quad (10)$$

$$\gamma_{ll}(l, t) = B_{v1}(l, t)^2 A_{\parallel}(l, t)^2, \quad (11)$$

$$\gamma_{\Omega\Omega}(l, t) = R_{v1}(l, t)^2 A_{\perp}(l, t)^2, \quad (12)$$

$$\beta^l(l, t) = -\dot{X}_{\text{coord}}(t)E(X(t))W(l)^{p_\beta}S(l - X(t)). \quad (13)$$

The capacity factors are anisotropic,

$$A_{\parallel} = \exp(qW \ln C_0), \quad (14)$$

$$A_{\perp} = \exp(qW \ln C_{\perp}), \quad (15)$$

$$T_{\text{pkt}} = \exp(qW \ln(\lambda C_0)). \quad (16)$$

The packet speed is measured against the stretched throat infrastructure:

$$\dot{X}_{\text{coord}} = \frac{U}{B_{\text{v1}}(X)}. \quad (17)$$

This is a service-coordinate statement. The throat's radial stretch remains useful, and packet motion must account for it.

The passenger packet is the finite region

$$|l - X(t)| \leq r_{\text{pkt}}. \quad (18)$$

The moving packet norm is

$$n_{\text{pkt}} = -\alpha^2 + \gamma_l(\dot{X}_{\text{coord}} + \beta^l)^2. \quad (19)$$

The stationary infrastructure monitor is

$$g_{tt} = -\alpha^2 + \gamma_l(\beta^l)^2. \quad (20)$$

The distinction between Eq. (19) and Eq. (20) is a major conceptual result. The packet norm tracks the passenger worldtube. Stationary g_{tt} tracks passive access and infrastructure state.

5 Diagnostics: measuring the burden channels

The reduced ADM diagnostic computes the extrinsic curvature

$$K_{ij} = \frac{D_i \beta_j + D_j \beta_i - \partial_t \gamma_{ij}}{2\alpha}, \quad (21)$$

the Hamiltonian source-demand proxy

$$\rho_H = \frac{{}^{(3)}R + K^2 - K_{ij}K^{ij}}{16\pi}, \quad (22)$$

and the momentum/current source-demand proxy

$$j_M^i = \frac{D_j(K^{ij} - \gamma^{ij}K)}{8\pi}. \quad (23)$$

The diagnostic groups these quantities into two channels:

$$J_{\text{dyn}} : \quad j_M \text{ and } K, \quad (24)$$

$$J_{\text{geom}} : \quad \rho_H \text{ and } {}^{(3)}R. \quad (25)$$

The conservative source-exposure proxy is

$$Tkk_{\text{floor}} = \rho_H - 2|j_M|. \quad (26)$$

A pressure-sensitive bracket adds a pressure-scale term,

$$Tkk_{\eta} = \rho_H + \eta \rho_H - 2|j_M|, \quad \eta \in \{-1, 0, +1\}. \quad (27)$$

These screens are engineering diagnostics. They tell a source-construction program where the candidate asks for support.

6 Result 1: passenger safety is packet-local

The first composite result is a change in what is being protected. In a passive-tunnel reading, one tries to make the whole active throat feel like a corridor. In the packet reading, the passenger is associated with $n_{\text{pkt}} < 0$ along the finite moving worldtube. The infrastructure can carry time-dependent support and transition burden outside that worldtube.

This separation explains several otherwise confusing reduced-screen outcomes. A stationary observer at a support edge can see a different diagnostic from the moving packet. A low-lapse infrastructure state can trigger passive access monitors while the service packet remains timelike under correct choreography. The design target is therefore packet safety plus controlled infrastructure burden.

This result also clarifies how the combined geometry uses the wormhole v1 system. The hybrid throat supplies plant functions: radial stretch, flare support, shoulder matching, repayment/reset, and sampled-stress ledgers. The passenger rides the packet.

7 Result 2: choreography is the hard packet gate

The second result is operational. The packet is protected by the order of events. The service layer must catch the packet before the shift fades, and the throat support must relax after the packet has cleared the critical transition:

$$x_{\text{catch}} < x_{\beta} < x_q. \quad (28)$$

The tested thresholds show the structure. With $x_{\beta} = 0.70$ and $x_q = 1.25$, higher-speed/lower-margin cases require earlier catch. At $V = 10$ and $\lambda = 6$, catch that begins too late creates packet failure; edge shaping then operates on a packet that has already lost the support sequence it needed.

Table 2: Catch threshold relative to shift fade. Higher service speed and lower lapse margin require earlier catch.

V	λ	first packet-fail x_{catch}	offset from x_{β}	first edge g_{tt} trigger
5	3	0.65	-0.05	—
5	5	0.70	0.00	—
10	6	0.60	-0.10	0.90
10	11.5	0.60	-0.10	—

The result identifies the kind of control that matters. The service sequence is the hard gate. Once the packet is caught early, medium-width profiles and edge shaping reduce the transition burden. When catch is late, edge smoothing arrives after the missing service order has already shaped the packet history.

8 Result 3: the transition layer carries current burden

The v01 transition refinement fixes the service order and then tunes the shape of catch, shift fade, throat relaxation, and edge taper. The selected transition layer is

$$x_{\text{catch}} = 0.05, \quad x_{\beta} = 0.70, \quad x_q = 1.25, \quad w_{\text{catch}} = 0.25, \quad w_{\beta} = 0.28, \quad w_q = 0.30, \quad p_{\beta} = 4. \quad (29)$$

Early catch gives the packet time to slow before shift fade. Medium transition widths reduce gradients while keeping the service phases distinct. The W^4 shift taper cleans the support edge after the packet sequence is correct.

The transition layer primarily moves the dynamic/current channel. In the stressed screen, the early-catch medium $p_\beta = 4$ transition reduces packet j_M and packet K to roughly $0.60\times$ the transition baseline, while strengthening packet timelike margins. Table 3 records the main readout.

Table 3: Transition refinement effect. The transition layer reduces the dynamic/current channel while improving packet margin.

Metric	Baseline	v01 transition	Relative
Packet max norm	-0.194	-0.704	stronger margin
Release max packet norm	-0.194	-1.819	stronger margin
Max packet j_M proxy	0.0397	0.0239	0.603
Max packet K proxy	13.73	8.25	0.601
Total source-cost proxy	1.000	0.939	0.939

The design implication is direct:

$$\boxed{\text{transition profiles are current and extrinsic-curvature controls.}} \quad (30)$$

Once this channel is reduced, the remaining large burden belongs to the spatial geometry.

9 Result 4: the spatial burden wants mixed capacity

The geometry refinement answers the next question: which support/capacity geometry reduces ρ_H and $^{(3)}R$ while preserving packet and edge clearance? The answer is mixed radial/angular capacity with broadened support.

The frozen balanced default is

$$\boxed{C_0 = 20, \quad C_\perp = 5, \quad R_{\text{th}} = 1.25, \quad w_{\text{th}} = 0.12, \quad R_{\text{pass}} = 0.35, \quad w_{\text{pass}} = 0.08, \quad B_0 = 6, \quad w_B = 12.} \quad (31)$$

The three-scenario confirmation grid uses

$$(V, \lambda) \in \{(5, 5.75), (10, 6), (10, 11.5)\}, \quad X \in \{0.05, 0.35, 0.70, 1.00, 1.25\}. \quad (32)$$

Table 4: Geometry confirmation grid. Costs are relative to the v01 geometry baseline.

Candidate	Total cost	Geometry cost	Dynamic cost	Packet max norm
v01 geometry baseline	1.000	1.000	1.000	-0.696
No-angular $C_\perp = 1$	0.574	1.000	0.00208	-0.762
$C_\perp = 3$ branch	0.174	0.298	0.00776	-0.762
$C_\perp = 5$ balanced default	0.162	0.274	0.0108	-0.762
Reserve $C_0 = 100, C_\perp = 5$	0.174	0.278	0.0341	-0.766

The no-angular comparison is the key control. With $C_\perp = 1$, the dynamic channel is tiny, but the geometry channel remains at baseline. With $C_\perp = 5$, the geometry channel falls to about $0.274\times$ baseline while the packet and support edge remain clear. Figure 2 shows the separation.

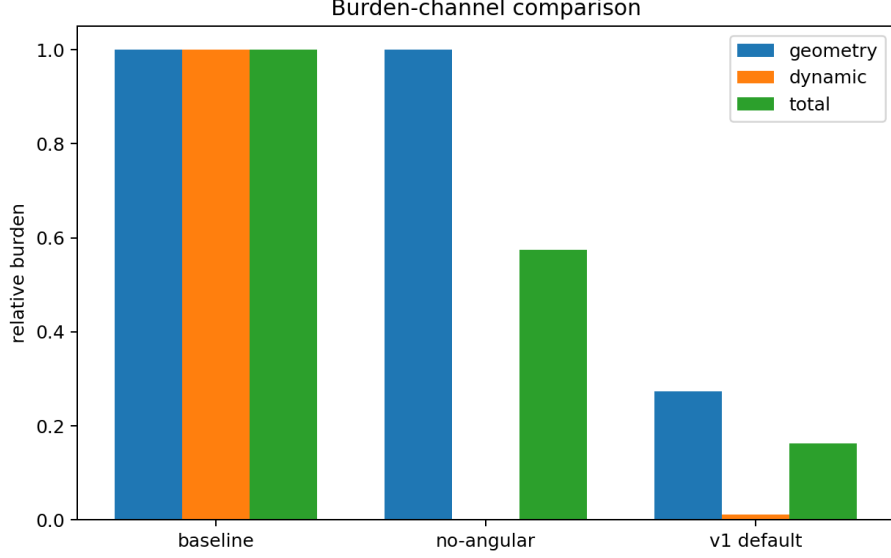


Figure 1: Burden-channel comparison. The mixed-capacity candidate reduces the geometry and dynamic channels relative to the v01 geometry baseline.

This is the core capacity result:

$$\boxed{\text{minimum useful radial capacity} + \text{moderate angular capacity.}} \quad (33)$$

The angular sector is a curvature-management actuator. It is part of the support plant.

10 Source exposure: the packet channel improves more than the total score predicts

The pre-freeze source screen applies the conservative current-floor proxy in Eq. (26). The v1 candidate improves the packet exposure strongly in the stressed $V = 10$, $\lambda = 6$ case.

Table 5: Conservative packet/source screen at $V = 10$, $\lambda = 6$.

Quantity	v01 baseline	v1 candidate	Relative
Packet min Tkk_{floor}	-0.789658	-0.124632	0.158
Packet negative-volume proxy	0.123668	0.011353	0.092
Packet j_M p95	0.300257	0.003341	0.011
Packet ρ_H p95	0.315277	0.096020	0.305
Packet $^{(3)}R$ p95	15.847573	4.826470	0.305
Packet K p95	67.616640	0.198121	0.003
Support-edge negative-volume proxy	0.024978	0.010403	0.416

The Lorentzian-sampled packet floor at $\tau = 0.2$ changes from

$$-2.0645 \times 10^{-2} \quad \text{to} \quad -8.704 \times 10^{-4}, \quad (34)$$

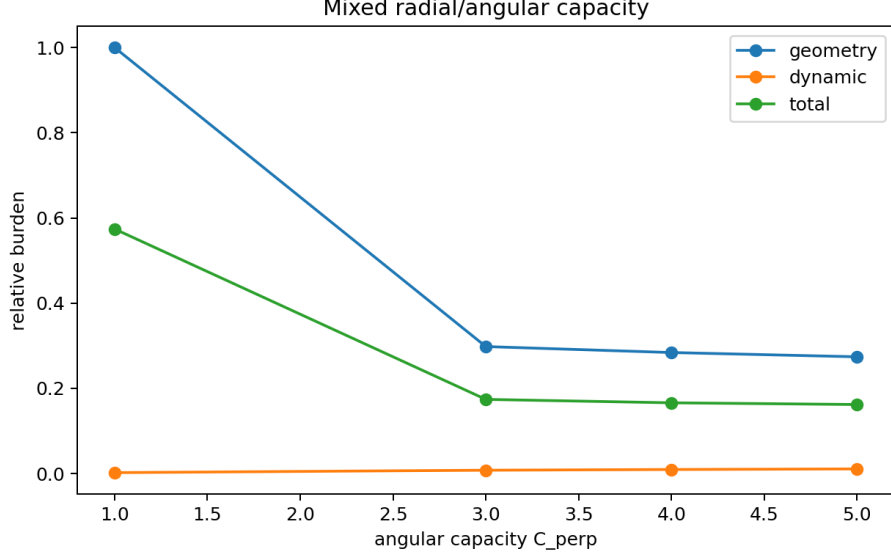


Figure 2: Mixed radial/angular capacity. The angular sector reduces the spatial-curvature burden that remains in a purely radial capacity design.

about $0.042\times$ the v01 baseline. The total geometry confirmation gave a much larger ratio, about $0.162\times$. The sampled packet exposure falls farther because the packet current channel is the channel most strongly suppressed by the transition-and-capacity combination.

This is a qualitative result and a numerical improvement. The architecture is doing the thing it was built to do: it cleans the packet channel. The infrastructure remains the place where support modeling has to happen. The passenger packet becomes the cleaner worldtube.

11 Robustness: a mixed-capacity operating band

The robustness checks refine the freeze. Pressure-sensitive source brackets, score-weighted rankings, and grid sensitivity all preserve the mixed-capacity family.

At 61×25 in the stressed case, $C_{\perp} = 5$ gives the best balanced total/geometry source-demand posture. The nearby $C_{\perp} = 3$ branch gives lower pressure/current exposure in some brackets. The design is therefore an operating band with nearby pressure/current margin.

Table 6: Pressure-sensitive robustness screen. Ratios are relative to the v01 baseline.

Design	Total	Geometry	Dynamic	floor exposure	pressure exposure
$C_{\perp} = 5$ balanced default	0.119	0.158	0.020	0.338	0.213
$C_{\perp} = 3$ pressure branch	0.142	0.192	0.015	0.283	0.144
$C_{\perp} = 1$ no-angular	0.721	1.000	0.005	1.000	0.026

The grid test preserves the ordering

$$\text{mixed-capacity candidate} < \text{no-angular comparison} < \text{v01 baseline} \quad (35)$$

for total cost across 25×11 , 41×17 , and 61×25 grids. Figure 4 shows that the ratios move with resolution while the design ordering remains stable.

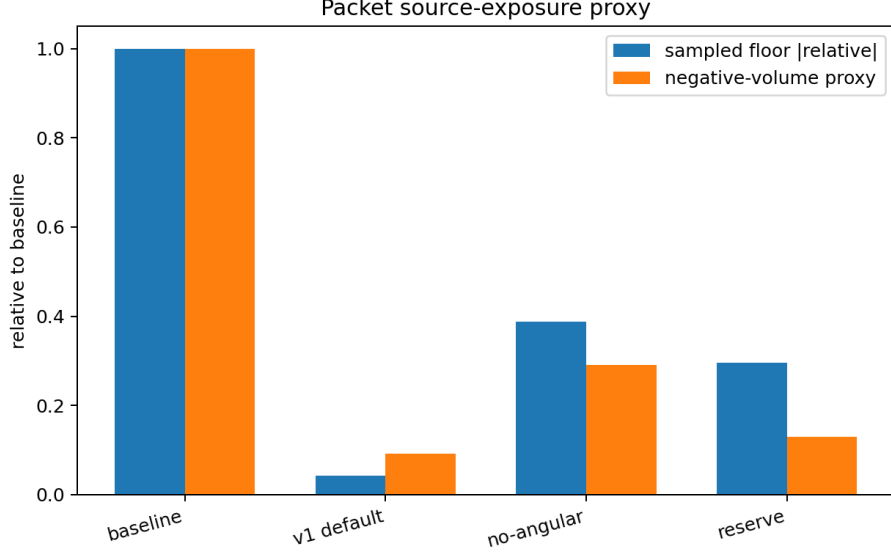


Figure 3: Packet source-exposure proxy at $V = 10$, $\lambda = 6$. The mixed-capacity candidate reduces sampled packet exposure and packet negative-volume proxy relative to the v01 baseline.

The freeze statement is therefore:

$$\boxed{\text{v1 balanced default: } C_0 = 20, C_{\perp} = 5, R_{\text{th}} = 1.25, w_{\text{th}} = 0.12, B_0 = 6, w_B = 12.} \quad (36)$$

The documented nearby branch is $C_{\perp} = 3$ for pressure/current-leaning studies.

12 Implications for source-supplied FTL engineering

The work changes the engineering picture in five ways.

First, it makes the protected object explicit. The passenger rides a packet. The active throat supports the packet. That lets the design evaluate passenger safety where it belongs and evaluate plant burden where it belongs.

Second, it gives timing a mechanical role. Catch is the service operation that makes shift fade safe. The sequence catch, shift fade, throat relaxation is the control law of the packet service.

Third, it creates two source-demand channels. Transition profiles move j_M and K ; capacity/support geometry moves ρ_H and $^{(3)}R$. This is the map a source-construction program can use. It says which source component should be designed to carry current, and which source component should be designed to carry spatial curvature.

Fourth, it identifies angular capacity as real infrastructure. A purely radial service geometry can give low current and high curvature burden. Moderate angular capacity reduces the curvature burden. This is the strongest capacity-design lesson in the paper.

Fifth, it shows alignment between the geometry screen and the source-exposure screen. The frozen candidate reduces the aggregate source-demand proxy and also reduces sampled packet current-floor exposure. The packet channel improves more strongly than the total score alone suggests.

These implications are the practical contribution under the source-available assumption. The result organizes the hard source problem. It turns a single undifferentiated exotic-support burden into packet, transition, geometry, edge, and source-channel tasks.

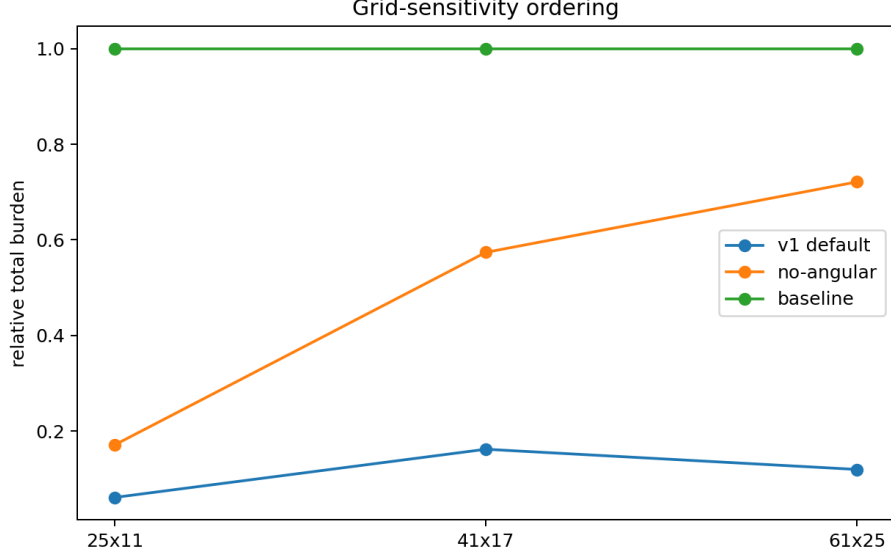


Figure 4: Grid-sensitivity ordering. The mixed-capacity family remains favored across the tested grids.

13 Next research gates

The frozen candidate supplies a clear target for the next phase. The next source task is pressure-complete source-channel construction,

$$T_{\mu\nu}^{\text{req}} = T_{\mu\nu}^{\text{support}} + T_{\mu\nu}^{\text{transition-current}} + T_{\mu\nu}^{\text{angular-capacity}} + T_{\mu\nu}^{\text{shoulder/matching}} + T_{\mu\nu}^{\text{packet-coupling}}. \quad (37)$$

The next relativity task is constraint-quality initial data for the frozen metric family. The next stability task is perturbation and repeated-cycle testing. The next global task is chronology-aware network scheduling. These gates now have a concrete candidate geometry and a burden-channel map.

14 Reproducibility package

The accompanying bundle contains LaTeX source, BibTeX references, compiled PDF, selected data, derived figures, source-access notes, and a checksum manifest. The data directory contains compact CSV outputs used in the paper. Full per-slice tensor archives can be regenerated from the diagnostic scripts in the larger project repository.

15 Conclusion

The Catch-Rematched Hybrid Flare-Gated Transit v1 candidate is a packet-centered active-rail spacetime architecture. The throat is infrastructure. The passenger is a protected packet. The shift is throat-gated. The service sequence is catch, shift fade, throat relaxation. The capacity geometry is mixed radial/angular. The burden map is separated into transition-current and spatial-curvature channels.

The frozen balanced default uses $C_0 = 20$, $C_\perp = 5$, broadened support, and moderate broad radial stretch. The reduced screens preserve packet and edge clearance, reduce the confirmed total source-demand proxy to about $0.162\times$ the v01 geometry baseline, and reduce Lorentzian-sampled packet negative exposure to about $0.042\times$ baseline in the stressed current-floor screen. Robustness checks identify a mixed-capacity operating band with $C_\perp = 5$ as the balanced default and $C_\perp = 3$ as a pressure/current-leaning branch.

The paper’s central result is a burden-distribution rule for source-supplied spacetime engineering: protect the packet, let the throat be plant, tune choreography for current, and tune angular capacity for curvature.

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